

WEB PROGRAMMING LAB-7 HOME EXERCISES

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Question 1

Develop a Web Application for Grocery Checklist Generation as shown in the figure below. It must have checkboxes which must be populated on page load listing grocery items. On clicking the Add Item button the selected Items and their prices have to be displayed in a Table. Set the borderstyle and border width for the table and its cells.

Create project & App

```
# Create Django project
django-admin startproject grocery-project
cd grocery-project

# Create app
python manage.py startapp groceryapp
```

grocery-project/settings.py

```
import os # added at the top

INSTALLED_APPS = [
    'groceryapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

Project URLs: `grocery-project/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('groceryapp.urls')),
]
```

App URLs: `groceryapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index, name='index'),
    path('result/', views.result, name='result'),
]
```

View: `groceryapp/views.py`

```
from django.shortcuts import render

# Page 1
def index(request):
    return render(request, 'index.html')

# Page 2 (handles form submission)
def result(request):
    if request.method == "POST":
        manufacturer = request.POST.get('manufacturer')
        model = request.POST.get('model')

        error = None

        # VALIDATION
        if not model or model.strip() == "":
            error = "Model name cannot be empty"

        if error:
            return render(request, 'index.html', {'error': error})
```

```

    context = {
      'manufacturer': manufacturer,
      'model': model.strip()
    }

    return render(request, 'result.html', context)

return render(request, 'index.html')

```

Create template/ in groceryapp/. Inside template/

grocery.html

```

<!DOCTYPE html>
<html>
<head>
  <title>Grocery Checklist</title>

  <style>
    table, th, td {
      border: 2px solid black;
      border-collapse: collapse;
      padding: 10px;
    }
  </style>

  <script>
    function addItem() {
      let items = document.getElementsByName("item");
      let table = document.getElementById("resultTable");

      // Clear old rows except header
      let rowCount = table.rows.length;
      for (let i = rowCount - 1; i > 0; i--) {
        table.deleteRow(i);
      }

      for (let i = 0; i < items.length; i++) {
        if (items[i].checked) {
          let row = table.insertRow();
          let cell1 = row.insertCell(0);
          let cell2 = row.insertCell(1);

          cell1.innerHTML = items[i].value;
          cell2.innerHTML = items[i].dataset.price;
        }
      }
    }
  </script>

```

```

</head>

<body>

<h2>Grocery Checklist</h2>

<!-- CHECKBOX LIST -->
<input type="checkbox" name="item" value="Rice" data-price="50"> Rice<br>
<input type="checkbox" name="item" value="Milk" data-price="30"> Milk<br>
<input type="checkbox" name="item" value="Bread" data-price="25"> Bread<br>
<input type="checkbox" name="item" value="Eggs" data-price="60"> Eggs<br>
<input type="checkbox" name="item" value="Fruits" data-price="80"> Fruits<br>

<br>

<button onclick="addItem()">Add Item</button>

<br><br>

<!-- TABLE -->
<table id="resultTable">
  <tr>
    <th>Item</th>
    <th>Price</th>
  </tr>
</table>

</body>
</html>

```

Run

```
python manage.py runserver 8080
```

Output

Grocery Checklist

- ☒ Rice
- ☐ Milk
- ☒ Bread
- ☐ Eggs
- ☒ Fruits

Add Item

Item	Price
Rice	50
Bread	25
Fruits	80